

Four-Core Pipelined Implied Volatility & Greeks Engine (Artix-7 200T @ 100 MHz)

AXI4-Stream Ingress  
(256-bit S, K, T, r, C\_mkt, TID)

Stage 1: Ingress  
Registration (1 Cycle)

- AXI4-Stream 256-bit Ingress: {S, K, T, r, C\_mkt, TID}
- Single-Cycle Register + TID Tag
  - No Ingress Divider
  - (Scale-Invariant Pad  Ratio eliminates 32-cyc divider)
- Q8.24 Fixed-Point Precision

Contract

Stage 2: Analytical Seeder  
(Brenner-Subrahmanyam, 64 Cyc)

$$\sigma_0 = \frac{2.5066}{\sqrt{T}} \cdot \frac{C_{mkt}}{(S+K)/2}$$

- Shared Digit-Recurrence  $\sqrt{T}$  (29 Cyc)
  - Forwards  $\sqrt{T}$  Directly to Stage 3
- Internal Non-Restoring Divider (33 Cyc)
  - Bounds Init Error:  $|\sigma_0 - \sigma^*| \leq 0.08$

$\sigma_0, \sqrt{T}$  (Fwd)

Stage 3: Core BS Datapath  
(126 Cycles, II = 1)

- Scale-Invariant Pad  Ratio (S-K)/(S+K)
- 18-Cyc CORDIC Log Fallback (SRL15-Matched)
- Horner 5th-Order Poly CDF N(d1), N(d2)
- Hyperbolic CORDIC exp(-rT) & exp(-d1<sup>2</sup>/2)
- Fully Unrolled 126-Stage Feedforward

$C_{BS}, \nu$   
(126 Cyc)

Stage 4: NR Update & Dividers  
(33 Cycles)

- Convergence:  $|C_{BS} - C_{mkt}| \leq \$0.01$
- u\_nr\_divider:  $\Delta\sigma = (C_{BS} - C_{mkt})/\nu$
- u\_gamma\_divider:  $\Gamma = \phi(d_1)/(S\sigma\sqrt{T})$
- Greeks:  $\Delta = N(d_1), \nu = S\sqrt{T}\phi(d_1)$
- TID-Scoreboard Priority Loopback

Converged (97.6%)

Stage 5: Multi-Core Arbiter  
& 128-Bit Egress (2 Cycles)

- Round-Robin 4-Core Complete Drain
- Packed 128-Bit Single-Flit Bus:
  - [127:122] TID Transaction ID (6b)
  - [121:96] Gamma Greek (26b)
  - [95:64] Vega Greek (32b)
  - [63:32] Delta Greek (32b)
  - [31:0] sigma Implied Vol (32b)

AXI4-Stream Egress  
(128-bit  $\sigma$  + Full Greeks)

Zero-BRAM Context Store

- 64 x 32-bit Distributed LUTRAM
- SRL32 Delay Shift Registers
- 0 Block RAMs (Zero-BRAM)
- Preserves 100% RAMB36/18 for 10GbE MAC & PCIe DMA

Context

Loopback  
 $\sigma_{n+1}$  (2.4%)